BMW SALES PERFORMANCE ANALYTICS DASHBOARD

Executive Summary: BMW Sales Dashboard Analysis

This project turns a CSV file of **50,000 BMW sales records (2010–2024)** into an interactive dashboard with filters (year, region, model, fuel type, sales classification), KPI cards, six Plotly charts, a data preview and CSV download. It exists as a Streamlit/Pandas/Plotly app (`app.py`) and a standalone HTML dashboard, plus a presentation and chart images. All figures in this report describe the supplied dataset only; its source is scribe and it is not verified official BMW data.

Overall Sales Performance

- Total sales volume is **253,375,734 units**, with an average listed price of **$75,035** per record.
- Average mileage is **100,307 km** and average engine size is **3.25 L**.
- **15,246 records (30.5%)** are classified High sales; they account for 51.1% of total volume.

Dataset Snapshot

Rows × columns	50,000 × 11 (no missing values, no duplicate rows)
Period / models / regions	2010–2024 / 11 models / 6 regions
Fuel types / transmissions	Diesel, Electric, Hybrid, Petrol / Automatic, Manual
Fields	Model, Year, Region, Color, Fuel_Type, Transmission, Engine_Size_L, Mileage_KM, Price_USD, Sales_Volume, Sales_Classification

Technologies

Python, Pandas (data processing), Streamlit (web app), Plotly / Plotly.js (charts), HTML/CSS/JavaScript (standalone dashboard), Matplotlib and python-pptx (static charts and presentation).

Yearly Sales Trend

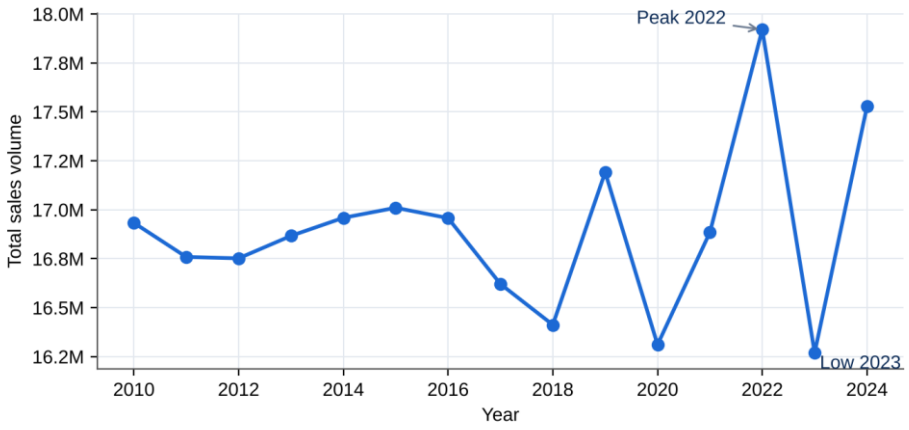


Figure 1: Total sales volume by year

- Volume stayed near 16.3M–17.2M units a year until 2021, then peaked at **17,920,946 in 2022**.
- It fell 9.2% to the low of **16,268,654 in 2023** and recovered to 17,527,854 in 2024 (3.5% above 2010).

Observation: Sales are stable overall but swing noticeably around 2022–2023.

Improvement Opportunity:

- Investigate the reasons for the 2022 peak and 2023 drop (launches, regional events or data-collection differences).
- Check the swings by region and model to see which segments drive them.

Sales by Region

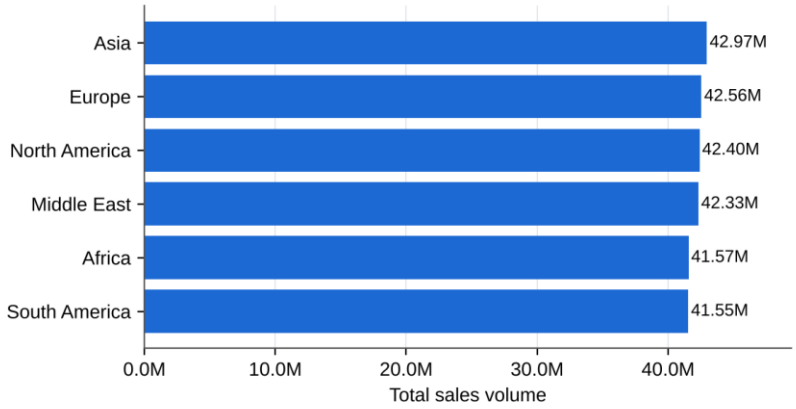


Figure 2: Total sales volume by region

- Asia leads with **42.97M units (17.0%)**, followed by Europe (16.8%) and North America (16.7%).

- Africa (16.4%) and South America (16.4%) are lowest; the top-to-bottom gap is only 3.4%.

Observation: Regional sales are almost evenly spread; no region dominates.

Improvement Opportunity:

- Study what helps Asia and Europe and test it in Africa and South America.
- Compare regional volume with market size to find unused potential.

Model Performance and Pricing

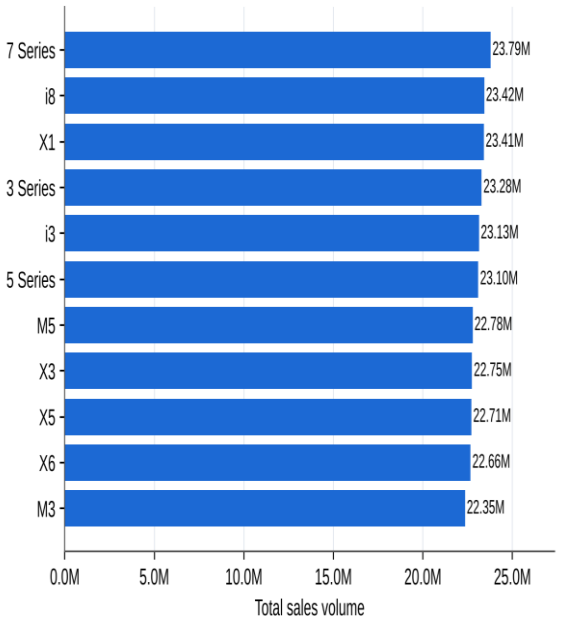


Figure 3: Sales volume by model

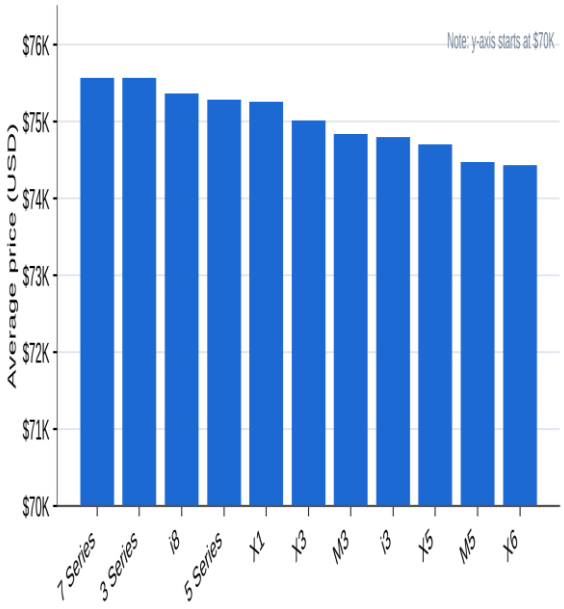


Figure 4: Average price by model (axis from $70K)

- **7 Series** leads (23.79M; 9.4%), then i8 (9.2%) and X1 (9.2%); **M3** is lowest (8.8%).
- Average price ranges from $75,570 (7 Series) to $74,435 (X6), only 1.5% apart.

Observation: Models differ very little in both volume and price.

Improvement Opportunity:

- Find why 7 Series, i8 and X1 sell slightly more and apply it to M3 and X6.
- Use region and fuel filters to see whether rankings change in specific markets.

Fuel Type and Transmission

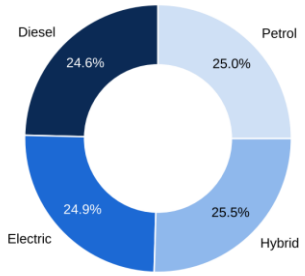


Figure 5: Sales volume share by fuel type

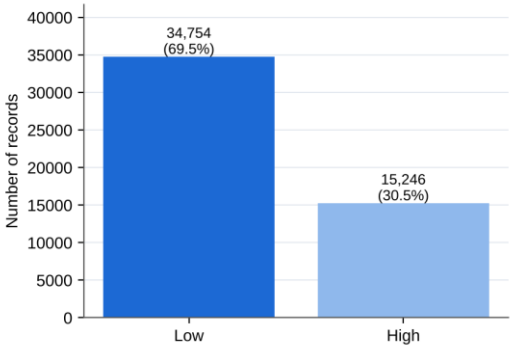


Figure 6: Records by sales classification

- **Hybrid** has the largest share (25.5%), then Petrol (25.0%), Electric (24.9%) and Diesel (24.6%). Manual (50.3%) and Automatic (49.7%) are almost equal.
- High-sales records are 30.5% of rows but 51.1% of volume (average 8,497 vs 3,563 units).

Observation: All four fuel types are within one percentage point of each other.

Improvement Opportunity:

- Track fuel-type share by year and region to expose trends hidden in the totals.
- Verify the data source before drawing conclusions about electrification.

Dashboard Overview

The interactive dashboard (Streamlit app and standalone HTML version) lets a user explore the dataset without writing any code. Figure 7 is the static preview supplied with the project.

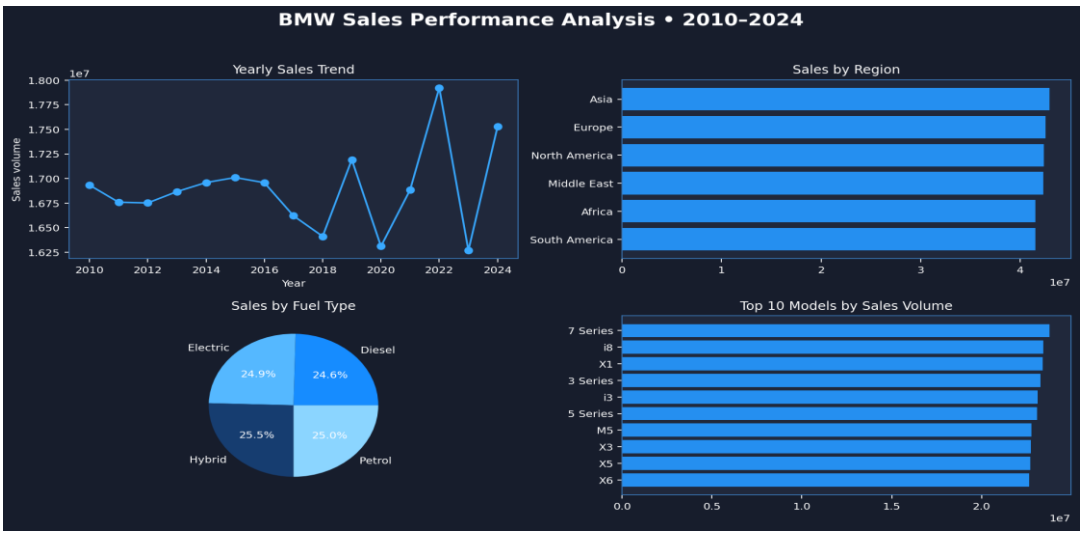


Figure 7: Dashboard preview

Dashboard Features

- **Filters:** Year, Region, Model, Fuel type and Sales classification (multi-select; all selected by default).
- **KPI cards:** total sales volume, number of records, average price, average mileage, and average engine size / high-sales records.
- **Charts:** yearly sales trend, sales by region, sales by fuel type, top models, average price by model, and price vs sales volume.
- **Data preview and download:** first 100 filtered rows are shown and all filtered rows can be downloaded as CSV.
- **Empty selection:** the app shows a warning (Streamlit) or an alert (HTML) instead of failing.

Technologies and Requirements

Python ≥ 3.10, Streamlit ≥ 1.36, Pandas ≥ 2.0, Plotly ≥ 5.20, python-pptx ≥ 0.6.23, Matplotlib ≥ 3.8; HTML/CSS/JavaScript with Plotly.js 2.35.2 for the standalone dashboard (needs internet to load the chart library); any modern browser. Run: `pip install -r requirements.txt`, then `python -m streamlit run app.py`.

Presentation

A six-slide PowerPoint (title, dataset overview, yearly trend, region and fuel type, model performance, takeaways) summarises the same analysis, and notes that findings describe only the supplied CSV.

Price versus Sales Volume

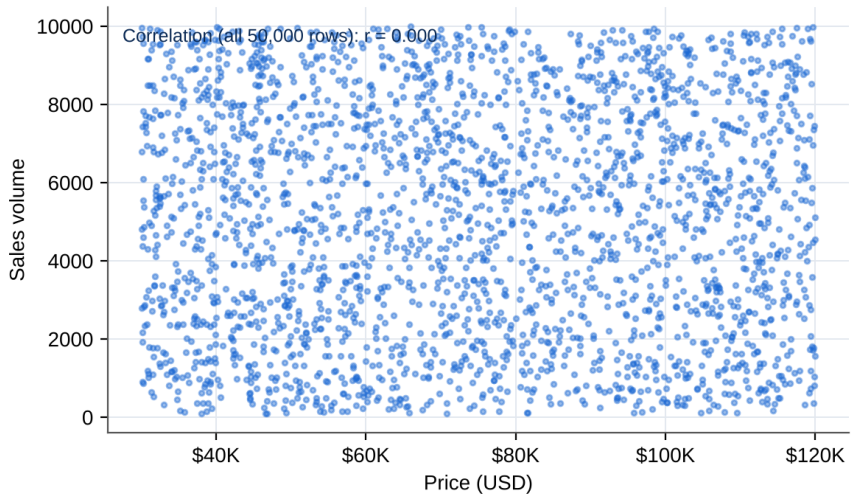


Figure 8: Price vs sales volume (random sample of 2,500 records)

Observation: Across all 50,000 records the correlation is r = 0.0001, i.e. price does not predict sales volume in this dataset (mileage and engine size also show no relationship).

Improvement Opportunity:

- Check how price and volume are defined in the source data before drawing business conclusions.
- Look at price and volume within a single model or region, where relationships may differ.

Testing Summary

Eighteen checks were run by script on the dataset and the filter logic of `app.py`; **all 18 passed**: 50,000 rows × 11 columns, no missing or duplicate values, totals and top region/model/fuel match `summary_stats.json`, default filters return all rows, a Year=2022/Asia/Electric filter returns exactly the 145 matching rows, and an empty selection returns no rows (warning shown). UI and performance tests were **not** performed. Live screenshots of the running app: [INSERT SCREENSHOTS].

Check	Expected	Result
Rows × columns	50,000 × 11	Pass
Missing / duplicate rows	0 / 0	Pass
Total sales volume vs summary_stats.json	253,375,734	Pass
Top region / model / fuel	Asia / 7 Series / Hybrid	Pass
Filter Year=2022, Asia, Electric	145 rows	Pass
Empty selection	0 rows → warning	Pass

Key Insights and Recommendations

1. **Sales are evenly distributed.** Asia, 7 Series and Hybrid lead (17.0%, 9.4%, 25.5%) but only narrowly; avoid over-reading small differences.
2. **Investigate 2022–2023.** The year-on-year swing is the clearest pattern in the data.
3. **Use filter combinations** (e.g. one region + one fuel type) where differences may be larger than in totals.
4. **Validate the dataset** (source, definitions of volume and price); price shows no link to sales, which is unusual for real market data.
5. **Add a screenshot gallery and refresh the data regularly** so that the dashboard stays current and easy to present.

Advantages, Limitations and Application

Advantages	Limitations
Interactive filters and instant KPIs/charts	Static CSV; no database or auto-refresh
Simple setup; free open-source tools	No login or user roles; HTML version needs internet for charts
Streamlit, HTML, PPTX and image outputs	Dataset origin undocumented
Downloadable filtered data	Descriptive analysis only; no forecasting

Applications: learning project for analytics and dashboards; template for sales dashboards in automotive and retail.

Future Enhancements (Proposed)

Database support for larger data; user login; forecasting and anomaly detection (for example for the 2022–2023 swing); automated tests; replacing the dataset with verified company data.

Conclusion

The dashboard successfully converts 50,000 sales records into an interactive, filterable analysis, and its data and filter logic passed all validation checks. Findings apply to the supplied dataset only.

References

[1] Streamlit docs: docs.streamlit.io. [2] Pandas docs: pandas.pydata.org. [3] Plotly docs: plotly.com/python and plotly.com/javascript. [4] Matplotlib: matplotlib.org. [5] python-pptx: python-pptx.readthedocs.io. [6] Dataset: scribe